

Akhilesh Kanmanthreddy

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EDUCATION

University of MichiganAnn Arbor, MI

B.S. Computer Science & Financial Mathematics, GPA: 3.8/4.0

Clubs & Involvement: Michigan Engineering Consulting Group, KTP Tech Fraternity, Youth Soccer Coach

Coursework: Data Structures and Algorithms, Discrete Math, Data Science, linear algebra, computer organization

EXPERIENCE

Nexteer AutomotiveMay 2026 - August 2026
Software Engineering InternSaginaw, MI

- Built **MCP servers** for IT Service Management (ITSM) and Production Control & Logistics (PC&L), exposing enterprise tools and data to AI agents to enable automated retrieval and interaction with internal systems
- Engineered a full-stack sales approval web application supporting **100+ users across 5 role-based user types** with authentication, approval workflows, comments, notifications, status tracking, search/filtering, and audit history
- Developed enterprise AI agents integrating **50 knowledge sources and tools**, enabling natural-language interaction with IT systems for **10,000 users** and reducing task completion time by approximately **60%**

Michigan Engineering Consulting Group (MECG)September 2025 – Present
Associate Project ManagerUniversity of Michigan — Ann Arbor, MI

Koops Automation – Technology & Data Engineering ConsultingJan 2026 – Present

- Designed and implemented a web-scraping system that monitored **20 public permit portals**, processing **100,000 records** and saving **175 hours of manual prospecting per week** across a 35-person sales team
- Built automated data pipelines to extract, clean, normalize, and evaluate **30 data fields per permit**, identifying **hundreds of commercially relevant opportunities** from large-scale public permit data
- Developed an automated alerting system that delivered qualified leads to the sales team same day, surfacing an estimated **\$5M+ in potential sales pipeline**

Red Hawk Golf Club – Financial & Marketing Strategy ConsultingSep 2025 – Dec 2025

- Conducted sensitivity analysis revealing break even utilization rate of **55%** at which open tee time revenue (**\$2,121**) **exceeds** fixed league revenue (**\$2,088**), informing a hybrid scheduling recommendation to maximize revenue
- Delivered strategic recommendations to company **CEO** across **3** areas (digital presence, staff training, and experience packaging) to **increase** tee sheet utilization during slow late morning to afternoon windows

PROJECTS/RESEARCH

Style | React Native, Expo, TypeScript, SupabaseFebruary 2026 – Present

- Built a full-stack iOS street wear discovery app aggregating **1M+ products** with profiles, saves, and personalized feeds.
- Engineered a semantic product search system using **FashionCLIP** to enable natural-language and visual similarity search
- Developed a personalized recommendation algorithm using **user swipe behavior, saved items, product attributes, and embedding similarity**

3D Modeling of Tumor Evolution | Python, TensorFlow, SKlearn, Pandas, NumPyAugust 2024 – May 2025

- Developed an LSTM based **3D** tumor growth forecasting system trained on sequential DICOM MRI scans, achieving **84.2%** volume overlap index, **5.84mm** Hausdorff distance, and **16.76%** volume error across **8** patients
- Automated tumor segmentation by training a Mask R-CNN on **500** manually labeled MRIs, predicting future anatomical brain invasion sites with **90%** accuracy via Yale Brain Atlas mapping
- Benchmarked LSTM, ARIMA, and exponential regression across **8** patients and **3** evaluation metrics, with LSTM outperforming all models by up to **79%** in volume accuracy

AeroSafe | C++, Arduino, MATLABAugust 2024 – March 2025

- Engineered the first **fully automated** pilot headset for hypoxia prevention, using Arduino sensors to detect low oxygen and autonomously deploy an oxygen mask, and secured **\$50,000** through startup competitions

NEUROX | Python, TensorFlow, Keras, OpenCVAugust 2022 – March 2023

- Engineered an optimal CNN architecture by systematically benchmarking **4** filter dimensions, **6** activation functions, **3** filter patterns, and **5** optimizers, achieving **98.65%** binary classification accuracy on **3,000** brain MRIs
- Trained a Mask R-CNN segmentation model on **40** manually labeled MRIs, achieving **71.88%** average tumor surface area accuracy via grid test evaluation

TECHNICAL SKILLS/ HOBBIES

Languages/Frameworks: Python, C++, HTML/CSS, Java, React Native, TypeScript,WordPress, Django, Expo

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Jupyter, Google Colab, Unity, Expo, Firebase, Firestore

Libraries: Pandas, NumPy, Matplotlib, TensorFlow, sklearn, PyGame, OpenAI, ElevenLabs, Difflib, Wpilib, Keras

Hobbies: Pickle ball, Basketball 3 point sharp shooter, Soccer, Aspiring master chef contestant